

Finite Nonviability Certificates under Delayed Information

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Abstract

We establish a continuous-to-finite theorem for robust nonviability under exogenous partial observation. An outer moment approximation of all measurable information-adapted controls, combined with an inner adversarial approximation, yields finite linear programs whose optimal values sandwich the continuous-time safety value between explicit error estimates. Every nonviable initial information state consequently admits a finite, independently verifiable obstruction certificate, and finite termination holds for every nonviable instance in the stated class. The certificates account for post-observation recourse: they can detect failure even when every compatible state is viable under full information and a common control keeps every branch safe throughout the blind window. Certificate sparsity is governed by information–time rank, and a matching construction shows that instantaneous input dimension cannot bound the number of necessary witnesses. A three-branch delayed-observation instance is carried exactly — delay threshold, Farkas certificate, and mesh refinement — and the finite programs are solved by a standard solver, with optimality certified in exact rational arithmetic by independent primal and dual witnesses.

1. Introduction

Viability theory under full observation has a mature sufficiency theory; under incomplete observation, the complementary question is when failure can be *certified* — finite, checkable witnesses that no observation-based policy is viable. The companion paper (Abaee, 2026) develops that calculus of obstruction certificates for discrete and continuous systems and names the failure mode its discrete certificates do not reach: a system can remain safe throughout the blind window and every compatible state can be viable under full information, yet the branches that could not be distinguished may have accumulated obligations that no post-observation control can discharge. This paper supplies the continuous-time instrument for exactly that mode: finite linear programs whose infeasibility certifies failure of *every measurable information-adapted controller*, not merely of sampled or parameterized controllers, with an explicit approximation-error bound, finite termination whenever nonviability holds, and a sparsity law governed by information–time rank.

The distinction that carries the paper is that the certifi-

cate may use safety obligations *after* information arrives: it prices the controller’s best post-observation recourse. It can therefore prove nonviability when every compatible state is individually viable under full information, when a common control keeps every branch safe throughout the blind interval, and when instantaneous admissibility and tangency tests find no obstruction.

Several ingredients are standard and are not advertised as new: robust safety as a family of linear inequalities, nonanticipativity constraints, an infinite-dimensional separating functional, Farkas’ lemma and Helly’s theorem, and monotonicity under observation refinement. The theorem-driven content is that one construction establishes, simultaneously: (i) the correct abstraction direction for nonviability — outer approximation of controller possibilities, inner approximation of adversarial tests; (ii) coverage of all measurable adapted controls, with no hidden zero-order-hold restriction; (iii) a computable two-sided error bound between a finite linear program and the continuous-time safety value; (iv) finite termination for every nonviable instance in the stated class; (v) a finite, policy-independent adversarial test set with numerical verification margins; and (vi) a sharp information–time sparsity law, including a construction showing that failure of any input-dimension-only bound.

Section 2 fixes the model and quantifier order. Sections 3–4 give the exact robust rows and the finite atomic certificate. Section 5 states the continuous-to-finite theorem. Section 6 carries a three-branch instance exactly and reports a solver campaign in which the finite programs are solved by HiGHS (Huangfu and Hall, 2018) and optimality is certified in exact rational arithmetic by independent witnesses. Section 7 records complexity and the certificate-verification rules. Section 8 establishes the sharp limitation that bounds by input dimension fail for blind control functions. Section 9 derives the refinement-erosion identity and the oracle-recourse extension. Section 10 delimits scope.

2. Model, information, and quantifiers

2.1 Dynamics

Let the hidden mode be $\theta \in \Theta := \{1, \dots, M\}$. Conditional on θ , the continuous state satisfies

$$\dot{x}(t) = A_\theta(t)x(t) + g_\theta(t) + B_\theta(t)u(t) + E_\theta(t)d(t), \quad t \in [0, T],$$

under the standing assumptions: $A_\theta, g_\theta, B_\theta, E_\theta$ bounded and piecewise continuous, with bounded-variation bounds on

B_θ, E_θ ; the initial set in mode θ a nonempty compact polytope $P_\theta \subset \mathbb{R}^n$; the input and disturbance sets nonempty compact polytopes $U = \{v : Fv \leq f\} \subset \mathbb{R}^m$ and $D_\theta \subset \mathbb{R}^p$; safety the closed polyhedron $\mathcal{V} = \{x : c_i^\top x \leq b_i, i = 1, \dots, s\}$; and the initial information state $B_0 = \bigcup_\theta \{\theta\} \times P_\theta$ initially safe (a time-zero certificate suffices otherwise). The dynamics admit unique Carathéodory solutions for every admissible measurable signal pair.

2.2 Information structure

At known times $0 = \tau_0 < \tau_1 < \dots < \tau_R = T$ the controller receives *exogenous* information about θ : refining history partitions $\mathcal{P}_0 \preceq \mathcal{P}_1 \preceq \dots \preceq \mathcal{P}_{R-1}$ of Θ , the observation history during $I_r = [\tau_r, \tau_{r+1})$ identifying the cell $C_r(\theta) \in \mathcal{P}_r$. These are history partitions, not necessarily partitions generated by the latest observation alone; nonrefining raw observations are replaced by their history refinement. Observations do not depend on x, u, d : active sensing and endogenous observation branching are excluded from the exact theorem (Section 9.2 gives a sound extension).

2.3 Policy class and quantifier order

Policies are deterministic, measurable, nonanticipative, and adapted to the observation history; controls may vary arbitrarily measurably between observations. Because each partition cell has a unique ancestor history, every such policy is represented exactly by signals $u_{r,C} \in L^\infty(I_r; U)$, $C \in \mathcal{P}_r$, with $\nu_\theta^\pi(t) = u_{r,C_r(\theta)}(t)$ on I_r . The robust quantifier order is

$$\exists \pi \in \Pi_{\mathcal{I}} \forall \theta \forall x_0 \in P_\theta \forall d \in L^\infty([0, T]; D_\theta) \forall t \in [0, T],$$

the adversary selecting mode, initial state, and measurable disturbance with knowledge of the announced policy; randomized policies are not included. The information-constrained viability kernel is

$$\mathcal{K}_{\mathcal{I}}^T = \left\{ B_0 : \exists \pi \forall \theta, x_0, d, t \quad x_{\theta, x_0, d}^\pi(t) \in \mathcal{V} \right\},$$

and a finite-horizon nonviability certificate also excludes infinite-horizon viability.

2.4 Relation to full-information viability

Let K_{full}^T be the physical-state viability kernel when (θ, x) is observed continuously, and define its belief lifting by $\text{Lift}(K_{\text{full}}^T) = \{B : \emptyset \neq B \subseteq K_{\text{full}}^T\}$. Then

$$\mathcal{K}_{\mathcal{I}}^T \subseteq \text{Lift}(K_{\text{full}}^T),$$

since a full-information controller can emulate any information-constrained policy. The inclusion is strict in general: the instance of Section 6 has every two-state subbelief viable while the three-state belief is not.

3. Adjoint safety rows and the safety value

Let $X_\theta(t, s)$ denote the state-transition matrix. For a label $a = (\theta, i, t) \in \mathcal{S} := \Theta \times \{1, \dots, s\} \times [0, T]$ define the input kernel and margin

$$k_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} B_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i,$$

$$\beta_a = b_i - h_{P_\theta}(X_\theta(t, 0)^\top c_i) - \int_0^t c_i^\top X_\theta(t, \sigma) g_\theta(\sigma) d\sigma - \int_0^t h_{D_\theta}(E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i) d\sigma$$

where $h_Q(v) = \sup_{q \in Q} v^\top q$. For a policy π set $\mathcal{F}_a(\pi) = \int_0^T k_a(\sigma)^\top u_\theta^\pi(\sigma) d\sigma - \beta_a$.

Proposition 1 (robust rows). π is robustly safe on $[0, T]$ if and only if $\mathcal{F}_a(\pi) \leq 0$ for every $a \in \mathcal{S}$. This follows from the linear solution formula; initial-state and disturbance maximizations separate from the control term because the control signal in a mode does not depend on x_0 or d . For polytopic D_θ , a maximizing disturbance can be chosen measurably by selecting a maximizing vertex at each time.

Define the minimax safety violation $J_{\mathcal{I}}(T) = \min_{\pi \in \Pi_{\mathcal{I}}} \max_{a \in \mathcal{S}} \mathcal{F}_a(\pi)$.

Proposition 2 (value and nonviability). The minimum exists: the policy-signal space is weak-* compact and each $\mathcal{F}_a$ is weak-* continuous. Consequently

$$B_0 \notin \mathcal{K}_{\mathcal{I}}^T \iff J_{\mathcal{I}}(T) > 0.$$

The strict positive margin is a consequence of compactness and closed safety constraints, not an extra nonviability assumption.

4. The finite obstruction certificate

Define $\phi_U(g) := \min_{v \in U} g^\top v = -h_U(-g)$. Choose finitely many labels $a_\ell = (\theta_\ell, i_\ell, t_\ell)$, $\ell = 1, \dots, q$, and weights $\lambda_\ell \geq 0$, $\sum_\ell \lambda_\ell = 1$. For each active information cell $C \in \mathcal{P}_r$, pool the adjoint kernels $g_{r,C}(\sigma) = \sum_{\ell: \theta_\ell \in C} \lambda_\ell k_{a_\ell}(\sigma)$. The certificate margin is

$$\Gamma_{\mathcal{I}}(\lambda, a) = \sum_{r=0}^{R-1} \int_{I_r} \sum_{C \in \mathcal{P}_r} \phi_U(g_{r,C}(\sigma)) d\sigma - \sum_{\ell=1}^q \lambda_\ell \beta_{a_\ell}. \quad (1)$$

Remark 1 (semantics). If $\Gamma_{\mathcal{I}}(\lambda, a) > 0$, then no admissible information-adapted policy is safe: for every policy, $\sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a)$, so at least one stored label satisfies $\mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a) > 0$ — an admissible realization violates facet i_ℓ at time t_ℓ . The violating label may depend on the policy; the finite list and its weights do not. The certificate is not an adversarial tree: it is a finite, policy-independent set of adversarial tests combined by an adjoint identity.

5. The continuous-to-finite theorem

The exact dual expression (1) alone would be close to standard infinite-dimensional convex duality. The consequential result is a sound finite construction covering unrestricted measurable controls.

5.1 Outer control moments

Choose a control-time mesh $\mathcal{H} = \{J_1, \dots, J_H\}$ containing every sensing time, so each J_j lies within one information stage. Introduce one vector $v_{j,C} \in U$ for every cell C active during J_j , and let $Q = \sum_j |\mathcal{P}_{r(j)}|$ be the number of information-time blocks. For a label $a = (\theta, i, t)$ define the finite row $A_{a,j,C} = \mathbf{1}_{\{\theta \in C\}} \int_{J_j} k_a(\sigma) d\sigma$, the block average $\bar{k}_{a,\mathcal{H}}(\sigma) = \frac{1}{|J_j|} \int_{J_j} k_a(s) ds$ on J_j , and the moment error

$$e_{a,\mathcal{H}} = \int_0^T h_U(\bar{k}_{a,\mathcal{H}}(\sigma) - k_a(\sigma)) d\sigma, \quad (2)$$

for which any verified upper bound $\bar{e}_a \geq e_{a,\mathcal{H}}$ may be used. For every measurable policy, its block averages $v_{j,C} = \frac{1}{|J_j|} \int_{J_j} u_{r(j),C}(\sigma) d\sigma$ belong to U , and

$$A_a v - \int_0^T k_a(\sigma)^\top u_\theta^\pi(\sigma) d\sigma \leq \bar{e}_a. \quad (3)$$

Inequality (3) is the crucial orientation: the finite moments yield a *necessary relaxation* of continuous-time safety. They do not restrict the real controller to held inputs. (For the sampled-data subclass whose policies are held constant on the blocks, the same polyhedron is an exact reduction rather than a relaxation — the held class is the policy class, so only inter-sample times are relaxed, which is sound.)

5.2 Inner adversarial approximation

One may use a right-hand side $\beta_a^+ \geq \beta_a$ obtained from an explicit admissible initial state and disturbance signal rather than exact worst-case optimization. If $0 \leq \beta_a^+ - \beta_a \leq \delta_\beta$ uniformly over the labels used, the approximation is safe for nonviability: it weakens the adversary and enlarges the feasible controller set. For example, with $w_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i$, on a disturbance mesh choose $d_{a,j} \in \arg \max_{d \in D_\theta} (\int_J w_a)^\top d$, yielding an explicit piecewise-constant admissible disturbance; if D_θ has radius R_D about a fixed center and $\text{TV}(w_a) \leq V_D$, then

$$0 \leq \beta_a^+ - \beta_a \leq R_D V_D h_d, \quad (4)$$

apart from any separately accounted initial-state or numerical error.

5.3 The finite program

Let $G \subset [0, T]$ be a finite time grid including every mode and safety facet at every grid time. Solve

$$\rho_{\mathcal{H},G} = \min_{v,\rho} \rho \quad \text{s.t.} \quad A_a v - \beta_a^+ - \bar{e}_a \leq \rho \quad (a \in \Theta \times \{1, \dots, s\} \times G), \quad (5)$$

Theorem 1 (continuous-to-finite). Assume: uniformly over admissible policies, modes, and facets, $|\mathcal{F}_{\theta,i,t}(\pi) - \mathcal{F}_{\theta,i,t'}(\pi)| \leq L|t - t'|$ (6); the input kernels satisfy $\text{TV}(k_a) \leq V_U$ (7); U has radius R_U about some fixed center; and the time grid has covering radius at most h_t , the control mesh maximum length h_u , and the adversarial approximation satisfies the bound above. Then:

(i) *Certified value sandwich.*

$$\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H},G} + R_U V_U h_u + \delta_\beta + L h_t. \quad (8)$$

With (4), $J_{\mathcal{I}}(T) - \rho_{\mathcal{H},G} \leq R_U V_U h_u + R_D V_D h_d + L h_t$. Verified numerical row widths are added to the right side.

(ii) *Sound finite certificates.* If $\rho_{\mathcal{H},G} > 0$, the program's dual supplies a finite certificate of nonviability of the original continuous-time system, covering every measurable information-adapted policy.

(iii) *Completeness under refinement.* If $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$, then every sufficiently fine sequence of the above meshes, with numerical errors tending to zero, eventually produces a positive finite certificate. Nonviability is thus semidecidable in this class when the coefficients and error bounds are effectively computable.

(iv) *Exact atomic dual representation.*

$$J_{\mathcal{I}}(T) = \sup_{q < \infty, a_1, \dots, a_q, \lambda \in \Delta_q} \Gamma_{\mathcal{I}}(\lambda, a). \quad (10)$$

In particular, nonviability is equivalent to the existence of a positive finite atomic certificate.

(v) *Sparse witnesses.* Let $r = \text{rank } A$ for the finite safety-row matrix in (5). Whenever $\rho_{\mathcal{H},G} > 0$, a positive certificate exists using at most $r+1 \leq mQ+1$ safety labels, and it can retain any prescribed margin strictly below $\rho_{\mathcal{H},G}$. If the lifted continuous-time kernels span a finite-dimensional space of dimension $r_{\mathcal{I}}$, a continuous-time obstruction has at most $r_{\mathcal{I}} + 1$ labels. The relevant dimension is information–time rank, not m .

The complete proof is in Supplementary S1. The asymmetry at its core is worth stating plainly: averaging arbitrary controls proves the lower bound; implementing piecewise-constant controls proves the upper bound. Confusing these two directions is what makes naïve infeasibility-by-discretization arguments unsound.

6. A three-branch delayed-observation instance

The instance is analytically checkable, polyhedral, and genuinely three-way: every pair of compatible states is viable, but the triple is not.

6.1 Plant and exact delay threshold

Consider the planar double integrator $\dot{p} = v$, $\dot{v} = u$, with unit vectors $n_1 = (1, 0)$, $n_2 = (-\frac{3}{5}, \frac{4}{5})$, $n_3 = (-\frac{3}{5}, -\frac{4}{5})$, the hexagonal input set $U = \text{conv}\{\pm n_1, \pm n_2, \pm n_3\}$, equivalently $\{\nu : |\nu_2| \leq \frac{4}{5}, |2u_1 + u_2| \leq 2, |2u_1 - u_2| \leq 2\}$, safety $n_i^\top p \leq 2$ ($i = 1, 2, 3$) and $|v_1|, |v_2| \leq \frac{6}{5}$, three possible initial states $p_j(0) = \frac{34}{25} n_j$, $v_j(0) = n_j$, no disturbance, and the mode unobserved until time τ , when it is revealed exactly. Every initial state is strictly inside the physical safety set.

If j is known, $u = -n_j$ for one second and then $u = 0$ keeps the maximum critical position at $\frac{34}{25} + \frac{1}{2} = \frac{93}{50} = 1.86 < 2$: all three physical states are indefinitely viable under full information. Take three witness labels, each using its own position facet at $t_* = \tau + 1$, with kernels $k_j(s) = \mathbf{1}_{\{s \leq t_*\}}(t_* - s)n_j$ and $\beta_j \frac{16}{5} \frac{1}{C} \frac{1}{25} U.t_*$, weighted by $\lambda_1 = \frac{3}{8}$, $\lambda_2 = \lambda_3 = \frac{5}{16}$, so that $\sum_j \lambda_j n_j = 0$. Before observation the pooled kernel is identically zero; after observation each mode has its own control and $\phi_U(\lambda_j(t_* - s)n_j) = -\lambda_j(t_* - s)$. Hence

$$\Gamma(\tau) = - \int_{\tau}^{\tau+1} (\tau + 1 - s) ds = - \left(\frac{16}{25} - \tau - 1 \right) = \tau - \frac{7}{50}, \quad (16)$$

so $\tau > \frac{7}{50}$ implies $B_0 \notin \mathcal{K}_{\mathcal{I}}$. Conversely, $u = 0$ before observation and $u = -n_j$ for one second afterward keeps the maximum critical position at $\frac{34}{25} + \tau + \frac{1}{2}$, safe exactly when $\tau \leq 0.14$. Thus $\tau_{\max} = \frac{7}{50} = 0.14$ is exact, and for horizons containing the stopping time $J_{\mathcal{I}}(\tau + 1) = \tau - 0.14$.

6.2 Pairs are viable; the blind-window exit test misses

At $\tau = \frac{1}{5}$, the common control $u = 0$ keeps all three branches safe throughout the blind window: $n_j^\top p_j(0.2) = \frac{39}{25} = 1.56 < 2$. No exit occurs before information arrives; nevertheless $\Gamma(0.2) = 0.06 > 0$ — the best post-observation recourse cannot recover all branches. The obstruction is not that information arrives after unavoidable physical exit; it is that information arrives after some indistinguishable branch has necessarily accumulated more braking liability than the remaining control authority can discharge.

Every pair remains viable. For $\{1, 2\}$ use the common blind control $w = (-\frac{2}{5}, -\frac{4}{5})$; reflect the second component for $\{1, 3\}$; for $\{2, 3\}$ use $w = (1, 0)$; after revelation brake the normal velocity with $-n_j$, then the remaining tangential velocity. With $\alpha = -n_j^\top w$, the maximum critical position is $\frac{34}{25} + \tau - \alpha\tau^2/2 + (1 - \alpha\tau)^2/2$, giving 1.9752 for $\{1, 2\}$, $\{1, 3\}$ ($\alpha = 0.4$) and 1.9352 for $\{2, 3\}$ ($\alpha = 0.6$), both below 2; all values are exact rationals (12345/6250 and 12193/6250). Over the seven nonempty priors formed from the three states, exactly six are viable at delay $\frac{1}{5}$ — a calculation over a prior family, not a claim that all seven are reachable beliefs from one fixed initial record.

6.3 The finite program and its Farkas certificate

Set $\tau = \frac{1}{5}$, $T = \frac{6}{5}$, $h = \frac{1}{10}$: two common pre-observation control blocks and ten post-observation blocks for each of three modes, so $Q = 32$, $mQ = 64$. For the endpoint position rows the exact moment error is $e = 12 \cdot h^2/4 = \frac{3}{100}$ (the kernel is affine on each of the $T/h = 12$ mesh intervals and $h_U(\alpha n_j) = |\alpha|$); the endpoint right-hand side is $\beta = \frac{16}{25} - \frac{6}{5} = -\frac{14}{25}$, relaxed

to $\beta + e = -\frac{53}{100}$. The exact continuous certificate has margin $\frac{3}{50} = 0.06$; the finite moment certificate has margin $\frac{3}{100} = 0.03$.

The dual witness is exact. Order the input facets by $F = [(0, 1); (0, -1); (2, 1); (2, -1); (-2, 1); (-2, -1)]$, $f = (\frac{4}{5}, \frac{4}{5}, 2, 2, 2, 2)$, and define $\eta_1 = (0, 0, 0, 0, \frac{1}{4}, \frac{1}{4})$, $\eta_2 = (0, \frac{1}{2}, 0, \frac{3}{10}, 0, 0)$, $\eta_3 = (\frac{1}{2}, 0, \frac{3}{10}, 0, 0, 0)$, satisfying $F^\top \eta_j = -n_j$ and $f^\top \eta_j = 1$. For a post-observation interval J_k put $a_k = \int_{J_k} (T - s) ds$ and $\mu_{j,k} = \lambda_j a_k \eta_j$. Before observation, stationarity holds because $\sum_j \lambda_j n_j = 0$; after observation, because $\lambda_j a_k n_j + F^\top \mu_{j,k} = 0$. Since $\sum_k a_k = \frac{1}{2}$, the Farkas contradiction is $\lambda^\top (\beta + e) + \mu^\top f_{\text{blk}} = -\frac{53}{100} + \frac{1}{2} = -\frac{3}{100} < 0$ — an exact rational certificate. The primal witness is equally exact: hold the pre-observation blocks at 0 and the post-observation blocks of mode j at $-n_j$; the endpoint row then reads $-\frac{1}{2} + \frac{14}{25} - \frac{3}{100} = \frac{3}{100}$, and every other row lies at or below the same value (verified row by row in exact arithmetic).

6.4 Solver-certified mesh study

The joint audit of the predecessor material left one verification step unexecuted: the mesh values below were known to match a closed-form lower bound, but LP optimality had not been certified by an actual solver run. That step is closed here. For each mesh the program (5) is built at its claimed dimensions and solved by HiGHS; optimality is then certified in exact rational arithmetic by the witness pair above (primal feasibility of every row; dual stationarity and dual objective $-\rho(h)$), so the duality gap closes exactly and the solver’s returned value is a computational confirmation of a proved optimum, not an observation. Table 1 reports the campaign: exact values $\rho(h) = \max\{0, \frac{3}{50} - Th/4\}$, solver values, iteration counts, and wall-clock times as measured on the verification machine.

The values obey the exact law $\rho(h) = \frac{3}{50} - Th/4$ for $h < \frac{1}{5}$ and vanish at $h = \frac{1}{5}$: the whole refinement gap is the control-moment error $Th/4$ of the endpoint rows. All dimension counts ($mQ + 1$ variables; $Ms(N_t + 1) + p_U Q$ inequalities), all witness inequalities, and the closed-form law are machine-verified in exact rational arithmetic by the companion verification script, which regenerates Table 1 verbatim and re-runs the solver campaign.

7. Computation, complexity, and independent verification

With Q information–time blocks and $N_t + 1$ time labels, the program has $mQ + 1$ variables and at most $Ms(N_t + 1) + p_U Q$ inequalities (p_U input-polytope facets); it is polynomial-time in these dimensions and the rational data size, conditional on the validated coefficient bounds. Initial-state dependence is affine, so no vertex enumeration of P_θ is required: at each label solve $\max_{\xi \in P_\theta} c_i^\top X_\theta(t, 0)\xi$ and store a maximizing or near-maximizing point — the verifier needs only membership of the stored point in P_θ , not global optimality; near-optimality bears on the convergence guarantee, not on a positive certificate’s validity. The same holds for stored disturbance signals.

For rational outer rows $\hat{A}v \leq \hat{b}$, $F_{\text{blk}}v \leq f_{\text{blk}}$, a residual-bounded certificate test suffices: with $\lambda, \mu \geq 0$, $r = \hat{A}^\top \lambda + F_{\text{blk}}^\top \mu$, $c = \lambda^\top \hat{b} + \mu^\top f_{\text{blk}}$, the condition

$$c + \sum_j h_U(-r_j) < 0 \quad (15)$$

is sufficient, since any feasible v satisfies $-\sum_j h_U(-r_j) \leq r^\top v \leq c$. All integration, rounding, and model-enclosure errors must be included before applying the test, and uncertain row data must be relaxed toward a *larger* feasible controller set: overestimating the worst disturbance contribution can create a false obstruction. A deletion-based sparsification uses at most

one feasibility solve per safety row; exact rank compression requires verified rank, and floating-point singular-value truncation must be charged to the error budget. If $J_{\mathcal{I}}(T) = \gamma > 0$, a mesh with $R_U V_U h_u + R_D V_D h_d + L h_t < \gamma$ guarantees detection — an explicit, though not claimed optimal, margin-dependent certificate-size bound.

8. Information–time rank: input dimension does not bound the witnesses

Proposition 3 (arbitrarily large minimal obstructions with one input). For every integer $q \geq 1$ there is a continuous-time system with scalar input $u(t) \in [0, 1]$ and $q + 1$ indistinguishable modes such that every proper subbelief is viable, the full $(q+1)$ -mode belief is not, and every obstruction certificate must involve all $q + 1$ modes.

Let $T = q$, $I_j = [j-1, j]$. For mode $j \leq q$: $\dot{x}_j = \mathbf{1}_{I_j}(t)(u-1)$, $x_j(0) = 0$; for mode $q+1$: $\dot{x}_{q+1} = -u$, $x_{q+1}(0) = q - \varepsilon$, $0 < \varepsilon < 1$. Require $0 \leq x \leq q$; there are no observations. Writing $w_j = \int_{I_j} u$, safety requires $-w_j \leq -1$ for $j \leq q$ and $\sum_j w_j \leq q - \varepsilon$; summing gives $0 \leq -\varepsilon$, a contradiction. If mode $q+1$ is removed, $u \equiv 1$ is safe; if mode $j \leq q$ is removed, $u = 0$ on I_j and $u = 1$ elsewhere is safe. Equal weights give certificate margin $\varepsilon/(q+1)$; the kernel rank is q , and $q + 1$ witnesses are necessary.

A statement such as “at most $m + 1$ compatible states suffice to witness nonviability” is valid for convex common-action sets in $\mathbb{R}^m$ but false for a common blind control function: the correct dimension counts independent temporal and informational control decisions. This distinction is consequential for review intervals, delayed sensing, and certificate-size claims; its discrete counterpart is recorded as a scoping caveat in the companion paper.

9. Refinement erosion and oracle recourse

Suppose a coarse information cell C is split into finer cells $C_1, \dots, C_k$. Because $\phi_U(g_1 + \dots + g_k) \geq \sum_j \phi_U(g_j)$, the same witness satisfies $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$, with the exact difference

$$\int [\phi_U(\sum_j g_{C_j}) - \sum_j \phi_U(g_{C_j})] dt. \quad (13)$$

For $U = [-\bar{u}, \bar{u}]^m$ this is $\bar{u} \int [\sum_j \|g_{C_j}(t)\|_1 - \|\sum_j g_{C_j}(t)\|_1] dt$: the certificate identifies the precise time intervals on which indistinguishable branches demand conflicting control effort. On rational instances the identity is exactly checkable; the verification record confirms $\phi_U(g_1 + g_2) \geq \phi_U(g_1) + \phi_U(g_2)$ on 3,240 deterministic rational pairs for the hexagonal U of Section 6.

The exact theorem assumes exogenous mode observations. A useful sound extension does not: if finitely many admissible initial-state/disturbance scenarios are provably indistinguishable until τ , give the controller after τ *separate arbitrary control signals for every stored scenario* — an oracle relaxation of the true policy class. If

$$\int_0^\tau \phi_U(\sum_\ell \lambda_\ell k_\ell) dt + \sum_\ell \int_\tau^T \phi_U(\lambda_\ell k_\ell) dt - \sum_\ell \lambda_\ell \beta_\ell > 0, \quad (14)$$

with β_ℓ computed from the explicitly stored scenario, then no actual output-feedback policy can be safe. This covers arbitrary sensing after a known blackout; it is sound but generally incomplete. The three-branch instance realizing this recourse mode in the discrete setting is in the companion paper.

10. Scope delimitations

The exact theorem does not cover: control-dependent observations or active sensing; arbitrary continuous-state measure-

h	blocks Q	variables $mQ+1$	rows $Ms(N_t+1) + p_U Q$	exact $\rho(h)$	solver ρ	iterations	time (s)
0.20	16	33	243	0	0.000000000	36	0.024
0.10	32	65	465	$\frac{3}{100} = 0.030$	0.030000000	78	0.014
0.05	64	129	909	$\frac{19}{200} = 0.045$	0.045000000	141	0.043
0.02	160	321	2241	$\frac{27}{500} = 0.054$	0.054000000	501	0.252
0.01	320	641	4461	$\frac{57}{1000} = 0.057$	0.057000000	1062	1.015

Table 1: Solver-certified mesh study on the three-branch instance ($\tau = \frac{1}{5}$, $T = \frac{6}{5}$): program dimensions, exact optimal values, HiGHS-attained values (agreement to at most 4.2×10^{-17}), simplex iterations, and wall-clock time. Optimality of each exact value is certified by the exact primal–dual witness pair of Section 6.3, independent of the solver. The exact continuous value is 0.06; three safety labels support each positive certificate.

ments; nonlinear dynamics without additional validated enclosures; nonconvex actuator sets without a relaxation gap; rate-constrained actuators without modifying the approximation; or unrestricted infinite-horizon completeness. For general sensing after a blackout, the oracle-recourse certificate is sound but incomplete. For nonlinear systems, a uniform validated flow-error bound can be added to the row budgets, yielding a sound extension, not automatic completeness. Against the alternatives: full-information viability decides none of the information-constrained question; initial admissibility and tangency tests see no obstruction here; safety-before-first-observation tests are silent when a common safe blind control exists; exact finite safety games await a justified continuous-to-finite bridge, which is what Theorem 1 provides; and naïve held-control infeasibility excludes only the held-control class. No general-purpose computational-calculus claim is made: the theorem is a certificate generator for the stated class, with completeness only under refinement and semidecidability only with effectively computable bounds.

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Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification scripts `paper2c_lp_campaign.py` (solver-certified mesh campaign and exact witnesses; regenerates Table 1 verbatim) and `paper2_nonstandard_verification.py` (51-check exact-arithmetic record for the predecessor material) are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`). **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.